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Solid oxide fuel cell and manufacturing method of the same

Abstract

A solid oxide fuel cell includes a support of which a main component is a metal, and an anode supported by the support, wherein the anode includes a first oxide having electron conductivity, wherein the first oxide is perovskite type oxide expressed as a composition formula $ABO_{3.3}$, wherein “A” of the composition formula includes at least one of Ca, Sr, Ba and La, wherein “B” of the composition formula includes at least Cr.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based upon and claims the benefit of priority of the prior Japanese Patent Application No. 2019-184725, filed on Oct. 7, 2019, the entire contents of which are incorporated herein by reference.

FIELD

(2) A certain aspect of the present embodiments relates to a solid oxide fuel cell and a manufacturing method of the same.

BACKGROUND

(3) The solid oxide fuel cell includes a solid oxide electrolyte layer having oxygen ion conductivity, an anode, and a cathode. At the anode, oxygen ions passing through the solid oxide electrolyte layer from the cathode react with hydrogen contained in a fuel gas. This reaction generates electrical power. It is necessary for the anode to have electron conductivity. And so, there is disclosed a technology in which an anode includes LaTiO₃-based material having the electron conductivity (see Japanese Patent Application Publication No. 2012-33418 and Japanese Patent Application Publication No. 2009-541955).

SUMMARY

(4) There is a problem that solid oxide fuel cells made of ceramics are susceptible to thermal shock or mechanical shock. And so, it is thought that the fuel cells are supported by a metal member so as to high resistance against vibration, shock or rapid rate of rising and falling temperature. However, a metallic support is fired in strong reductive atmosphere in order to suppress oxidation of the metallic support. And so, developing of an anode which can be fired in the strong reductive atmosphere is requested.

(5) According to a first aspect of the present embodiments, there is provided a solid oxide fuel cell including: a support of which a main component is a metal; and an anode supported by the support, wherein the anode includes a first oxide having electron conductivity, wherein the first oxide is perovskite type oxide expressed as a composition formula ABO₃, wherein “A” of the composition formula includes at least one of Ca, Sr, Ba and La, wherein “B” of the composition formula includes at least Cr.

(6) According to a second aspect of the present embodiments, there is provided a manufacturing

method of a solid oxide fuel cell, the manufacturing method including: preparing a multilayer structure in which a support green sheet and an anode green sheet are stacked, the support green sheet including metal material powder, the anode green sheet including perovskite type oxide as ceramic material powder, the perovskite type oxide being expressed by a composition formula ABO_{3-x} in which “A” of the composition formula includes at least one of Ca, Sr, Ba and La and “B” of the composition formula includes at least Cr, firing the multilayer structure in reductive atmosphere.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic cross-sectional view of a multilayer structure of a fuel cell;
- (2) FIG. 2 is an enlarged cross-sectional view of details of a support, a mixed layer, and an anode; and
- (3) FIG. 3 is a flowchart of the manufacturing method of a fuel cell.

DETAILED DESCRIPTION

(4) Hereinafter, an embodiment will be described with reference to the accompanying drawings.

(5) FIG. 1 is a schematic cross-sectional view of a stacking structure of a solid oxide fuel cell **100**. As illustrated in FIG. 1, the fuel cell **100** has, as an example, a structure in which a mixed layer **20**, an anode **30**, an electrolyte layer **40**, a reaction prevention layer **50**, and a cathode **60** are stacked on a support **10** in this order. A plurality of the fuel cells **100** may be stacked to structure a fuel cell stack.

(6) The electrolyte layer **40** is a dense layer that is mainly composed of solid oxide having oxygen ion conductivity and has gas impermeability. The electrolyte layer **40** is preferably mainly composed of scandia yttria stabilized zirconium oxide (ScYSZ). The oxygen ion conductivity is the highest when the concentration of $Y_{2O_3}+Sc_{2O_3}$ is 6 mol % to 15 mol %. Thus, use of a material having this composition is preferable. The thickness of the electrolyte layer **40** is preferably 20 μm or less, further preferably 10 μm or less. The thinner electrolyte layer is better. However, to prevent gas at the upper face side and gas at the lower face side from leaking, the thickness is preferably 1 μm or greater.

(7) The cathode **60** is an electrode having electrode activity as a cathode, and has electron conductivity and oxygen ion conductivity. For example, the cathode **60** is lanthanum strontium cobaltite (LSC) having electron conductivity and oxygen ion conductivity. LSC is $LaCoO_3$ doped with strontium (Sr).

(8) The reaction prevention layer **50** is mainly composed of a component that prevents reaction between the electrolyte layer **40** and the cathode **60**. For example, the reaction prevention layer **50** is mainly composed of Gd-doped ceria (GDC) that is CeO_2 doped with gadolinium (Gd). As an example, when the electrolyte layer **40** contains ScYSZ and the cathode **60** contains LSC, the reaction prevention layer **50** prevents the following reactions.



(9) FIG. 2 is an enlarged cross-sectional view illustrating details of the support **10**, the mixed layer **20**, and the anode **30**. As illustrated in FIG. 2, the support **10** is a member that has gas permeability and is able to support the mixed layer **20**, the anode **30**, the electrolyte layer **40**, the reaction prevention layer **50**, and the cathode **60**. The support **10** is a porous metallic material, and is, for example, a porous material of Fe—Cr alloys.

(10) The anode **30** is an electrode having electrode activity as an anode, and has an electrode bone structure made of a ceramic material. The electrode bone structure contains no metallic component. In this configuration, decrease in the porosity in the anode due to coarsening of a metallic

component is inhibited during firing in a high-temperature reductive atmosphere. Additionally, alloying with a metallic component of the support **10** is inhibited, and deterioration of the catalyst function is inhibited.

(11) The electrode bone structure of the anode **30** preferably has electron conductivity and oxygen ion conductivity. The anode **30** preferably contains a first oxide **31** as an electron conductive material. The first oxide **31** can be a perovskite-type oxide expressed by the composition formula of ABO_{3-x} where the A site is at least one selected from a group consisting of Ca, Sr, Ba, and La, and the B site includes at least Cr. The perovskite-type oxide has favorable electron conductivity. The mole fraction of the B site may be equal to or greater than the mole fraction of the A site ($B \geq A$). More specifically, the first oxide **31** can be a LaCrO_{3-x} -based material. The B site of the perovskite-type oxide included in the first oxide **31** may include another element such as Ti, in addition to Cr. For example, when the LaCrO_{3-x} -based material is expressed by $\text{LaCr}_{1-x}\text{Ti}_x\text{O}_{3-x}$ because the B site includes both Ti and Cr, x is 0.5 or more and 1 or less.

(12) When the amount of CrO_{3-x} generated by decomposition of the first oxide **31** is large in the anode **30**, the anode **30** may not necessarily achieve favorable electron conductivity. And so, it is preferable that the amount of CrO_{3-x} has a lower limit. For example, when the cross section of the anode **30** is observed, it is preferable that {the area of CrO_{3-x} /(the area of CrO_{3-x} and the area of the first oxide **31**)} is 10% or less. It is more preferable that {the area of CrO_{3-x} /(the area of CrO_{3-x} and the area of the first oxide **31**)} is 5% or less. It is still more preferable that {the area of CrO_{3-x} /(the area of CrO_{3-x} and the area of the first oxide **30**)} is 1% or less. It is possible to measure the area of the perovskite oxide and the area of CrO_{3-x} , by SEM-EDS mapping of the cross section.

(13) The electrode bone structure of the anode **30** preferably contains a second oxide **32** as an oxygen ion conductive material. The second oxide **32** is ScYSZ or the like. For example, it is preferable to use ScYSZ having the following composition range. Scandia (Sc_2O_3) is 5 mol % to 16 mol %, and yttria (Y_2O_3) is 1 mol % to 3 mol %. It is more preferable to use ScYSZ of which the total additive amount of scandia and yttria is 6 mol % to 15 mol %. This is because the highest oxygen ion conductivity is obtained in this composition range. The oxygen ion conductive material is, for example, a material with a transference number of oxygen ion of 99% or greater. GDC may be used as the second oxide **32**. In the example of FIG. 2, a solid oxide identical to the solid oxide contained in the electrolyte layer **40** is used as the second oxide **32**.

(14) As illustrated in FIG. 2, in the anode **30**, for example, the first oxide **31** and the second oxide **32** form the electrode bone structure. This electrode bone structure forms a plurality of pores. A catalyst is carried on the surface exposed to the pore of the electrode bone structure. Thus, in the spatially continuously formed electrode bone structure, a plurality of catalysts is spatially dispersed. A composite catalyst is preferably used as the catalyst. For example, a third oxide **33** having oxygen ion conductivity and a catalyst metal **34** are preferably carried, as a composite catalyst, on the surface of the electrode bone structure. The third oxide **33** may be, for example, $\text{BaCe}_{1-x}\text{Zr}_x\text{O}_{3-x}$ doped with Y (BCZY, x=0 to 1), $\text{SrCe}_{1-x}\text{Zr}_x\text{O}_{3-x}$ doped with Y (SCZY, x=0 to 1), LaScO_3 doped with Sr (LSS), or GDC. Ni or the like may be used as the catalyst metal **34**. The third oxide **33** may have a composition identical to that of the second oxide **32**, or may have a composition different from that of the second oxide **32**. A metal acting as the catalyst metal **34** may be in a form of compound when electric power is not generated. For example, Ni may be in a form of a nickel oxide (NiO). These compounds are reduced with a reductive fuel gas supplied to the anode **30**, and becomes in a form of metal acting as an anode catalyst.

(15) From a viewpoint of favorable power generation performance of the fuel cell **100**, it is preferable that the area of the electron conductive path, the area of the ion conductive path, and the area of the pore are appropriately balanced. For example, in the cross section of the anode **30**, it is

preferable that each of the area of the first oxide **31**, the area of the second oxide **32** and the area of the pore is 20% or more and 60% or less. For example, in the cross section of the anode **30**, the area ratio of the first oxide **31** is 60%, the area ratio of the second oxide **32** is 20%, and the area ratio of the pore is 20%. Alternatively, in the cross section of the anode **30**, it is preferable that each of the area ratio of the first oxide **31**, the area ratio of the second oxide **32**, and the area ratio of the pore is 33%.

(16) The mixed layer **20** contains a metallic material **21** and a ceramic material **22**. In the mixed layer **20**, the metallic material **21** and the ceramic material **22** are randomly mixed. Thus, a structure in which a layer of the metallic material **21** and a layer of the ceramic material **22** are stacked is not formed. Also in the mixed layer **20**, a plurality of pores is formed. The metallic material **21** is not particularly limited as long as the metallic material **21** is a metal. In the example of FIG. 2, a metallic material identical to the metallic material of the support **10** is used as the metallic material **21**. The first oxide **31** or the second oxide **32** can be used as the ceramic material **22**. For example, ScYSZ, GDC, a LaTiO.sub.3-based material, or a LaCrO.sub.3-based material can be used as the ceramic material **22**. Since the LaTiO.sub.3-based material and the LaCrO.sub.3-based material have high electron conductivity, the ohmic resistance in the mixed layer **20** can be reduced. It is preferable that the same oxide of the first oxide **31** is used as the ceramic material **22**.

(17) The fuel cell **100** generates power by the following actions. An oxidant gas containing oxygen, such as air, is supplied to the cathode **60**. At the cathode **60**, oxygen reaching the cathode **60** reacts with electrons supplied from an external electric circuit to become oxygen ions. The oxygen ions conduct through the electrolyte layer **40** to move to the anode **30** side. On the other hand, a fuel gas containing hydrogen, such as a hydrogen gas or a reformed gas, is supplied to the support **10**. The fuel gas reaches the anode **30** through the support **10** and the mixed layer **20**. Hydrogen reaching the anode **30** release electrons at the anode **30** and reacts with oxygen ions conducting through the electrolyte layer **40** from the cathode **60** side to become water (H.sub.2O). The released electrons are drawn out to the outside by the external electric circuit. The electrons drawn out to the outside are supplied to the cathode **60** after doing electric work. Through the above-described actions, electric power is generated.

(18) In the above-described power generation reaction, the catalyst metal **34** acts as a catalyst in the reaction between hydrogen and oxygen ions. The first oxide **31** conducts electrons obtained by the reaction between hydrogen and oxygen ions. The second oxide **32** conducts oxygen ions reaching the anode **30** from the electrolyte layer **40**.

(19) The fuel cell **100** of the embodiment has the support **10** of which the main component is a metal. Therefore, the fuel cell **100** has resistance against thermal shock, mechanical shock and so on. The support **10** can be fired in strong reductive atmosphere because the main component of the support **10** is a metal. The first oxide **31** is the perovskite type oxide of which the composition formula is expressed as ABO.sub.3. The A site includes at least one of Ca, Sr, Ba, and La. The B site includes at least Cr. It is possible to fire the anode **30** in the strong reductive atmosphere. It is therefore possible to fire the support **10** together with the anode **30**. And it is possible to make the fuel cell **100**. The first oxide **31** has favorable electron conductivity. Therefore, the fuel cell **100** can achieve favorable power generation performance.

(20) In the fuel cell **100** according to the present embodiment, the support **10** is mainly composed of a metal, and the electrode bone structure of the anode **30** is composed of ceramic. In the aforementioned structure, when firing is performed such that the support **10** and the anode **30** are in contact with each other, the difference in material properties between metal and ceramic may cause interlayer peeling between the support **10** and the anode **30**. In particular, ceramic fired in a strong reductive atmosphere has poor adhesiveness with metal. In contrast, it is preferable that the fuel cell **100** according to the present embodiment includes the mixed layer **20** interposed between the support **10** and the anode **30**. Since the mixed layer **20** contains the metallic material **21** and the ceramic material **22**, the mixed layer **20** has material properties of metal and material properties of

ceramic. Thus, the mixed layer **20** has high adhesiveness with the support **10** and high adhesiveness with the anode **30**. Therefore, interlayer peeling between the support **10** and the anode **30** can be inhibited.

(21) Additionally, in the fuel cell **100** according to the present embodiment, the third oxide **33** is carried on the electrode skeleton of the anode **30**. This structure allows to first form the electrode bone structure by firing, and then, impregnate the electrode bone structure with the third oxide **33** and fire the electrode bone structure at low temperature. Thus, even when the second oxide **32** and the third oxide **33** do not have the same composition, reaction between oxides is inhibited. Therefore, the degree of freedom to select, as the third oxide **33**, an oxide appropriate to the composite catalyst is increased.

(22) Additionally, the porosity in the support **10**, the porosity in the mixed layer **20**, and the porosity in the anode **30** preferably have a relationship of the porosity in the support **10**>the porosity in the mixed layer **20**>the porosity in the anode **30**. This relationship allows the support **10** to have sufficient gas permeability. Since the anode **30** has a comparatively low porosity, the anode **30** obtains high electron conductivity and high oxygen ion conductivity while maintaining gas permeability. The mixed layer **20** obtains gas permeability and the contact area with the support **10** is obtained. Thus, the mixed layer **20** obtains the adhesion with the support **10**. The porosity is calculated from the ratio of the estimated area of the pores to the entire area based on an SEM image of the cross-section of a sample.

(23) Hereinafter, a manufacturing method of the fuel cell **100** will be described. FIG. 3 is a flowchart of the manufacturing method of the fuel cell **100**.

(24) Making Process of Material for Support

(25) Metallic powder having a particle size of, for example, 10 μm to 100 μm , a plasticizer, a solvent, a vanishing material, and a binder are mixed to make slurry as a material for support. The amount of the plasticizer is adjusted to, for example, 1 wt % to 6 wt % to adjust the adhesiveness of the sheet. The solvent is toluene, 2-propanol (IPA), 1-butanol, terpineol, butyl acetate, ethanol, or the like. The amount of the solvent is 20 wt % to 30 wt % depending on the viscosity or the like. The vanishing material is an organic substance. The binder is PVB, acrylic resin, ethyl cellulose, or the like. The material for support is used as a material for forming the support **10**. The ratio of the volume of the organic components (the vanishing material, the solid component of the binder, and the plasticizer) to the volume of the metallic powder is within a range of, for example, 1:1 to 20:1. The amount of the organic components is adjusted depending on the porosity.

(26) Making Process of Material for Mixed Layer

(27) Ceramic material powder, which is the raw material of the ceramic material **22**, metallic material powder having a small particle size, which is the raw material of the metallic material **21**, a solvent, a plasticizer, and a binder are mixed to make slurry as a material for mixed layer. The ceramic material powder has a particle size of, for example, 100 nm to 10 μm . The metallic material powder has a particle size of, for example, 1 μm to 10 μm . The solvent is toluene, 2-propanol (IPA), 1-butanol, terpineol, butyl acetate, ethanol, or the like. The amount of the solvent is 20 wt % to 30 wt % depending on the viscosity. The amount of the plasticizer is adjusted to 1 wt % to 6 wt % to adjust the adhesiveness of the sheet. The vanishing material is an organic substance. The binder is PVB, acrylic resin, ethyl cellulose, or the like. The ratio of the volume of the organic components (the vanishing material, the solid component of the binder, and the plasticizer) to the volume of the ceramic material powder and the metallic material powder is within a range of, for example, 1:1 to 5:1. The amount of the organic components is adjusted depending on the porosity. The diameter of the pore is controlled by adjusting the particle size of the vanishing material. The ceramic material powder may contain powder of an electron conductive material and powder of an oxide-ion conductive material. In this case, the ratio of the volume of the powder of the electron conductive material to the volume of the powder of the oxide-ion conductive material is preferably within a range of, for example, 1:9 to 9:1. Use of an electrolyte material such as ScYSZ, GDC, or

the like instead of the electron conductive material also prevents the peeling of the boundary face and enables the manufacture of the cell. However, to reduce the ohmic resistance, it is preferable to mix an electron conductive material and metallic powder.

(28) Making Process of Material for Anode

(29) Ceramic material powder structuring the electrode bone structure, a solvent, a plasticizer, a vanishing material, and a binder are mixed to make slurry as a material for anode. The solvent is toluene, 2-propanol (IPA), 1-butanol, terpineol, butyl acetate, ethanol, or the like. The amount of the solvent is 20 wt % to 30 wt % depending on the viscosity. The amount of the plasticizer is adjusted to, for example, 1 wt % to 6 wt % to adjust the adhesiveness of the sheet. The vanishing material is an organic substance. The binder is PVB, acrylic resin, ethyl cellulose, or the like. Powder of the electron conductive material that is the raw material of the first oxide **31** and has a particle size of, for example, 100 nm to 10 μ m and powder of the oxygen ion conductive material that is the raw material of the second oxide **32** and has a particle size of, for example, 100 nm to 10 μ m may be used as the ceramic material powder structuring the electrode bone structure. The ratio of the volume of the organic components (the vanishing material, the solid component of the binder, and the plasticizer) to the volume of the powder of the electron conductive material is within a range of, for example, 1:1 to 5:1, and the amount of the organic components is adjusted depending on the porosity. Additionally, the diameter of the pore is controlled by adjusting the particle size of the vanishing material. The ratio of the volume of the powder of the electron conductive material to the volume of the powder of the oxygen ion conductive material is within a range of, for example, 3:7 to 7:3.

(30) Making Process of Material for Electrolyte Layer

(31) Powder of an oxygen-ion conductive material, a solvent, a plasticizer, and a binder are mixed to make slurry as a material for electrolyte layer. The powder of the oxygen-ion conductive material is, for example, ScYSZ, YSZ, GDC, or the like, and has a particle size of 10 nm to 1000 nm. The solvent is toluene, 2-propanol (IPA), 1-butanol, terpineol, butyl acetate, ethanol, or the like. The amount of the solvent is 20 wt % to 30 wt % depending on the viscosity. The amount of the plasticizer is adjusted to 1 wt % to 6 wt % to adjust the adhesiveness of the sheet. The binder is PVB, acrylic resin, ethyl cellulose, or the like. The ratio of the volume of the organic components (the solid component of the binder and the plasticizer) to the volume of the powder of the oxygen ion conductive material is within a range of, for example, 6:4 to 3:4.

(32) Making Process of Material for Cathode

(33) Powder of lanthanum strontium cobaltite (LSC: LaSrCoO.sub.3), a solvent, a plasticizer, and a binder are mixed to make slurry as a material for cathode. The solvent is toluene, 2-propanol (IPA), 1-butanol, terpineol, butyl acetate, ethanol, or the like, and the amount of the solvent is 20 wt % to 30 wt % depending on the viscosity. The plasticizer is adjusted to 1 wt % to 6 wt % to adjust the adhesiveness of the sheet. The binder is PVB, acrylic resin, ethyl cellulose, or the like. The ratio of the volume of the organic components (the solid component of the binder, the plasticizer) to the volume of LSC powder is within a range of, for example, 6:4 to 1:4.

(34) Firing Process

(35) A support green sheet is made by applying the material for support on a polyethylene terephthalate (PET) film. A mixed layer green sheet is made by applying the material for mixed layer on another PET film. An anode green sheet is made by applying the material for anode on yet another PET film. An electrolyte layer green sheet is made by applying the material for electrolyte layer on yet another PET film. For example, several support green sheets, one mixed layer green sheet, one anode green sheet, and one electrolyte layer green sheet are stacked in this order, cut into a predetermined size, and fired within a temperature range of approximately 1100° C. to 1300° C. in a reductive atmosphere with an oxygen partial pressure of 10.sup.-16 atm or less. Through the above process, a half cell including the support **10**, the mixed layer **20**, the electrode bone structure of the anode **30**, and the electrolyte layer **40** is obtained. The reductive gas flown into the furnace

may be a gas in which H.sub.2 (hydrogen) is diluted with non-flammable gas (Ar (argon), He (helium), N.sub.2 (nitrogen) or the like). 100% of the reductive gas may be H.sub.2. From a view point of safety such as explosion limit, it is preferable that the concentration of H.sub.2 has an upper limit. For example, it is preferable that the concentration of H.sub.2 is 4 volume % or less, when the reductive atmosphere is a mixed gas of H.sub.2 and Ar.

(36) Impregnating Process

(37) Next, the electrode bone structure of the anode **30** is impregnated with the raw materials of the third oxide **33** and the catalyst metal **34**. For example, the following process is repeated as many times as needed such that Gd-doped ceria or Sc, Y-doped zirconia and Ni are generated when the cell is fired in a reductive atmosphere at a predetermined temperature. Nitrate or chloride of Zr, Y, Sc, Ce, Gd, or Ni is dissolved in water or alcohol (ethanol, 2-propanol, methanol or the like). A half cell is impregnated with the resulting solution, and dried. The resulting half cell is subjected to heat treatment.

(38) Forming Process of Reaction Preventing Layer

(39) As the reaction prevention layer **50**, Ce.sub.0.8Gd.sub.0.2O.sub.2-x is formed so as to have a thickness of 1 μm by, for example, PVD.

(40) Forming Process of Cathode

(41) Next, the material for cathode is applied on the reaction prevention layer **50** by screen printing and is then dried. Then, a cathode is sintered by heat treatment. Through the above process, the fuel cell **100** can be fabricated.

(42) In the manufacturing method of the embodiment, the first oxide **31** is the perovskite type oxide of which the composition formula is expressed as ABO.sub.3. The A site includes at least one of Ca, Sr, Ba, and La. The B site includes at least Cr. It is possible to stably fire the anode **30** in the strong reductive atmosphere. It is therefore possible to fire the support **10** together with the anode **30**. And it is possible to make the fuel cell **100**. The main component of the support **10** is a metal. Therefore, the fuel cell **100** has resistance against thermal shock, mechanical shock and so on. The first oxide **31** has favorable electron conductivity. Therefore, the fuel cell **100** can achieve favorable power generation performance.

(43) When the oxygen partial pressure in the firing atmosphere is 10.sup.16 atm or less, it is possible to suppress the generation of CrO.sub.3 caused by the decomposition of the first oxide **31**. Therefore, the first oxide **31** achieves favorable electron conductivity. From a view point of suppression of the generation of CrO.sub.3, it is preferable that the oxygen partial pressure in the firing atmosphere is 10.sup.-20 atm or less. It is more preferable that the oxygen partial pressure is 10.sup.-22 atm or less.

(44) Since the material for mixed layer contains the metallic material powder and the ceramic material powder, the mixed layer **20** after the firing the metallic material **21** and the ceramic material **22**. Therefore, the mixed layer **20** has material properties of the metallic material **21** and the ceramic material **22**. Therefore, interlayer peeling between the support **10** and the anode **30** can be inhibited.

(45) Additionally, the amount of the vanishing material in the material for support, the amount of the vanishing material in the material for mixed layer, the amount of the vanishing material in the material for anode are preferably adjusted such that the porosity in the support **10**, the porosity in the mixed layer **20**, and the porosity in the anode **30** have a relationship of the porosity in the support **10**>the porosity in the mixed layer **20**>the porosity in the anode **30**. This relationship allows the support **10** to have sufficient gas permeability. The anode **30** becomes dense, and obtains high oxygen ion conductivity. The mixed layer **20** obtains gas permeability, and also obtains the contact area with the support **10**, thereby obtaining adhesion with the support **10**.

(46) In the manufacturing method according to the present embodiment, the electrode bone structure is first formed by firing. Then, the electrode bone structure is impregnated with the composite catalyst and fired at low temperature (for example, 850° C. or less). Thus, even when the

second oxide **32** and the third oxide **33** do not have the same composition, the reaction between oxides is inhibited. Therefore, the degree of freedom to select, as the third oxide **33**, an oxide appropriate to the composite catalyst is increased.

EXAMPLES

(47) The fuel cell **100** was fabricated according to the manufacturing method in accordance with the above embodiment.

Example 1

(48) SUS (stainless) powder was used as the material for support. ScYSZ was used as the electrolyte layer **40**. A LaCrO₃-based material was used for the first oxide **31** of the anode **30**, and ScYSZ was used for the second oxide **32**. GDC was used for the third oxide **33**, and Ni was used for the catalyst metal **34**. The electrode bone structure was formed of the first oxide **31** and the second oxide **32**. A LaCrO₃-based material was used for the ceramic material **22** of the mixed layer **20**. Stainless steel (SUS) was used for the metallic material **21** of the mixed layer **20**. The cell after stacking was fired under a reductive atmosphere with an oxygen partial pressure of 10⁻¹⁶ atm or less. Additionally, the resulting cell was impregnated with the composite catalyst, and was then fired under an air atmosphere at a temperature of 850° C. or less.

Example 2

(49) ScYSZ was used as the electrolyte layer **40**. A LaCr_{0.9}Ti_{0.1}O₃-based material was used for the first oxide **31** of the anode **30**, and ScYSZ was used for the second oxide **32**. GDC was used for the third oxide **33**, and Ni was used for the catalyst metal **34**. The electrode bone structure was formed of the first oxide **31** and the second oxide **32**. A LaCrO₃-based material was used for the ceramic material **22** of the mixed layer **20**. SUS was used for the metallic material **21** of the mixed layer **20**. The cell after stacking was fired under a reductive atmosphere with an oxygen partial pressure of 10⁻¹⁶ atm or less. Additionally, the resulting cell was impregnated with the composite catalyst, and was then fired under air atmosphere at a temperature of 850° C. or less.

Comparative Example 1

(50) A LaTiO₃-based material was used for the first oxide **31** of the anode **30**. A LaTiO₃-based material was used for the ceramic material **22** of the mixed layer **20**. Other conditions were the same as those of the example 1.

Comparative Example 2

(51) A LaCrO₃-based material was used for the material for support. No mixed layer **20** was provided. The cell after the stacking was fired under normal atmosphere. Other conditions were the same as those of example 1.

(52) Table 1 shows the manufacturing conditions of the examples 1 and 2, and the comparative examples 1 and 2.

(53) TABLE-US-00001

TABLE 1	ANODE BONE STRUCTURE	MIXED OHMIC REACTION	FIRST SECOND LAYER	RESISTANCE	RESISTANCE	ELECTROLYTE	OXIDE	OXIDE
OXIDE	METAL	SUPPORT (Ω .Math. cm.sup.2)	(Ω .Math. cm.sup.2)	EXAMPLE 1	ScYSZ	LaCrO ₃	ScYSZ	LaCrO ₃
LaCrO ₃	ScYSZ	LaCrO ₃	SUS	SUS	0.25	0.28	POWDER	POWDER
EXAMPLE 2	ScYSZ	LaCr _{0.9} Ti _{0.1} O ₃	ScYSZ	LaCrO ₃	SUS	SUS	0.26	0.28
POWDER	POWDER	COMPARATIVE	ScYSZ	LaTiO ₃	ScYSZ	LaTiO ₃	SUS	SUS
0.25	0.4	EXAMPLE 1	POWDER	POWDER	COMPARATIVE	ScYSZ	LaCrO ₃	ScYSZ
—	—	LaCrO ₃	0.31	0.78	EXAMPLE 2			

Evaluation of Electrical Power Generation

(54) Impedance of each of the fuel cells of the examples land 2 and the comparative examples 1 and 2 was measured to separate the resistance value to measure the ohmic resistance of the fuel cell as a whole and the reaction resistance of the anode **30**. In the example 1, the ohmic resistance was 0.25 Ω.Math.cm.sup.2, and the reaction resistance of the anode **30** was 0.28 Ω.Math.cm.sup.2. In the example 2, the ohmic resistance was 0.26 Ω.Math.cm.sup.2, and the reaction resistance of the

anode 30 was $0.28 \Omega \cdot \text{cm} \cdot \text{sup.2}$.

(55) From the results of the examples 1 and 2, it is thought that the anode 30 was stably fired in the strong reductive atmosphere, the first oxide 31 achieved favorable electron conductivity, and the ohmic resistance was favorable, because the first oxide 31 of the anode 30 was the perovskite type oxide expressed as the composition formula $\text{ABO}_{0.3}$ in which the A site included at least one of Ca, Sr, Ba and La, and the B site included at least Cr. From the result of the example 2, it is understood that the anode 30 achieves favorable electron conductivity even if another element such as Ti is added to the B site of the perovskite type oxide of the first oxide 31 in addition to Cr.

(56) The cross section of the cell of the example 1 after the firing was observed by the SEM-EDS. {the area of $\text{CrO}_{0.3}$ /(the area of $\text{CrO}_{0.3}$ and the area of the first oxide 31)} in the anode 30 was 5% or less. It is thought that this was because the anode 30 was fired in the strong reductive atmosphere and the flocculation of $\text{CrO}_{0.3}$ was suppressed. The area ratio of the first oxide 31 in the cross section of the anode 30 was 27%. The area ratio of the second oxide 32 was 27%. The area ratio of the pores was 46%. Interlayer peeling did not occur in the examples 1 and 2. It is thought that this was because the mixed layer 20 was provided.

(57) In the comparative example 1, the ohmic resistance was $0.25 \Omega \cdot \text{cm} \cdot \text{sup.2}$, and the reaction resistance of the anode was $0.40 \Omega \cdot \text{cm} \cdot \text{sup.2}$. In this manner, the electrical power generation performance of the comparative example 1 was less than that of the examples 1 and 2. In the strong reductive atmosphere, resistance of the $\text{LaTiO}_{0.3}$ -based material against reductive force is small, and oxygen defects occur because of reduction of Ti. In this case, grain growth tends to be promoted. The anode becomes dense more easily than the $\text{LaCrO}_{0.3}$ -based material of the examples 1 and 2. It is thought that the porosity of the anode after the firing was reduced, and the electrical power generation performance gets worse. The cross section of the anode was observed by the SEM-EDS. The area of the pores in the anode using the $\text{LaTiO}_{0.3}$ -based material was 30% or less in the whole of the anode. It is thought that this was because gas diffusion became worse, and the high performance of the electrical power generation of the example 1 was not achieved.

(58) In the comparative example 2, the ohmic resistance was $0.31 \Omega \cdot \text{cm} \cdot \text{sup.2}$, and the reaction resistance of the anode was $0.78 \Omega \cdot \text{cm} \cdot \text{sup.2}$. In this manner, the electrical power generation performance of the comparative example 2 was less than that of the examples 1 and 2. It is thought that this was because the resistance value of the ceramic support was larger than that of the metallic support. Moreover, the anode was fired in the normal atmosphere, the $\text{LaCrO}_{0.3}$ -based material of the first oxide 31 was decomposed, and a large amount of $\text{CrO}_{0.3}$ was generated. The cross section of the cell of the comparative example 2 after the firing was observed by the SEM-EDS. {the area of $\text{CrO}_{0.3}$ /(the area of $\text{CrO}_{0.3}$ and the area of the first oxide 31)} in the anode was approximately 30%.

(59) Although the embodiments of the present invention have been described in detail, the present invention is not limited to such a specific embodiment, and it is to be understood that the various change, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

Claims

1. A solid oxide fuel cell comprising: a support of which a main component is a metal; an anode supported by the support; and a mixed layer provided between the support and the anode, wherein the anode includes a first oxide which is $\text{LaCrO}_{0.3}$ -based perovskite of which an A site is only La and of which a B site is only Cr and Ti and a second oxide having oxygen ion conductivity, wherein the support, the mixed layer and the anode are sintered bodies which are obtained by arranging a powder material of a $\text{LaCrO}_{0.3}$ -based material and a powder metal material for the mixed layer mixed together with a width in a stacking direction, on a powder metal material for the

support, and arranging a powder material including the first oxide on the powder material of a LaCrO_{3-x} -based material and a powder metal material for the mixed layer, and sintering the powder metal material for the support, the powder material of a LaCrO_{3-x} -based material and the powder metal material for the mixed layer, and the powder material including the first oxide, wherein the LaCrO_{3-x} -based material of the mixed layer is LaCrO_{3-x} -based perovskite of which an A site is only La and of which a B site is only Cr or only Cr and Ti, wherein the anode forms an electrode bone structure with the first oxide and the second oxide having oxygen ion conductivity, wherein the bone structure contains no metallic component, is spatially continuously formed, and includes pores and catalysts, and wherein each of the catalysts is carried on a surface exposed to each of the pores, wherein an ohmic resistance of the solid oxide fuel cell as a whole is $0.26 \, \Omega \cdot \text{cm}^2$ or less, and wherein a reaction resistance of the anode is $0.28 \, \Omega \cdot \text{cm}^2$ or less.

2. The solid oxide fuel cell as claimed in claim 1, wherein {an area of CrO_3 /(the area of CrO_3 and an area of the first oxide)} is 10% or less, in a cross section of the anode.
 3. The solid oxide fuel cell as claimed in claim 1, wherein an area ratio of a total of the first oxide/a total of (the first oxide+the second oxide+the pores) in the cross section of the anode is 20% or more and 60% or less, wherein an area ratio of a total of the second oxide/the total of (the first oxide+the second oxide+the pores) in the cross section of the anode is 20% or more and 60% or less, and wherein an area ratio of a total of the pores/the total of (the first oxide+the second oxide+the pores) in the cross section of the anode is 20% or more and 60% or less.
 4. The solid oxide fuel cell as claimed in claim 1, wherein a porosity in the support, a porosity in the mixed layer, and a porosity in the anode have a relationship of the porosity in the support>the porosity in the mixed layer>the porosity in the anode.
 5. The solid oxide fuel cell as claimed in claim 1, wherein each of the catalysts includes a catalyst metal and a third oxide having oxygen ion conductivity.
 6. The solid oxide fuel cell as claimed in claim 1, wherein the second oxide is scandia yttria stabilized zirconium oxide.
 7. The solid oxide fuel cell as claimed in claim 1, wherein {an area of CrO_3 /(the area of CrO_3 and an area of the first oxide)} is 5% or less, in a cross section of the anode.
 8. The solid oxide fuel cell as claimed in claim 1, wherein the second oxide of the anode is scandia yttria stabilized zirconium oxide; and wherein the catalysts are Gd-doped ceria and Ni.
 9. The solid oxide fuel cell as claimed in claim 1, wherein the LaCrO_{3-x} -based material of the mixed layer is $\text{LaCr}_x\text{Ti}_{1-x}\text{O}_3$ and x is 0.5 or more and less than 1.
 10. The solid oxide fuel cell as claimed in claim 1, wherein an A site of the LaCrO_{3-x} -based material of the mixed layer is only La, and wherein a B site of the LaCrO_{3-x} -based material of the mixed layer is only Cr and Ti.
 11. The solid oxide fuel cell as claimed in claim 1, wherein the first oxide is $\text{LaCr}_x\text{Ti}_{1-x}\text{O}_3$ and x is 0.5 or more and less than 1.
 12. The solid oxide fuel cell as claimed in claim 1, wherein a ratio of the reaction resistance to the ohmic resistance is 1.12 or less.
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